

TASK-9

4-07-2026

TOPIC: AWS EC2 Networking Architecture
(VPC, Subnet, Route Table, and Internet Gateway)

Step 1: Create a VPC:

- Open AWS Console
- Go to VPC service.
- Click Create VPC.
- Name: my-vpc
- **IPv4 CIDR: 10.0.0.0/16**
- Click Create VPC.

vpc-076a0f9df0ce8dbe3 Act

Details Info

VPC ID  vpc-076a0f9df0ce8dbe3	State  Available	Block Public Access  Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-055e65bdf3946ed58	Main route table rtb-01ddf85ae298d82ee
Main network ACL acl-0e1e1591cc7aace95	Default VPC Yes	IPv4 CIDR 172.31.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID  964376925119
Encryption control ID -	Encryption control mode -		

Step 2: Create a Subnet:

- Go to VPC → **Subnets**.
- Enter Name: **public-subnet**.
- Enter IPv4 CIDR block: **10.0.1.0/24**.
- Click Create subnet.

Subnets (1/8) Info Last updated 2 minutes ago Actions Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC
☐	-	subnet-0414d6e5e989edac3	Available	vpc-076a0f9df0ce8dbe3
☐	-	subnet-0d6757bd3b9b4b5b8	Available	vpc-076a0f9df0ce8dbe3
☒	karthik	subnet-02cc3d0aee80d3f52	Available	vpc-0ad023c4fa4cd32c2 karthik

Step 3: Create an Internet Gateway:

- VPC → Internet Gateways.
- Click Create internet gateway.
- **Name: my-igw**
- Click Create.
- **Actions → Attach to VPC.**
- Choose my-vpc.
- Click Attach.

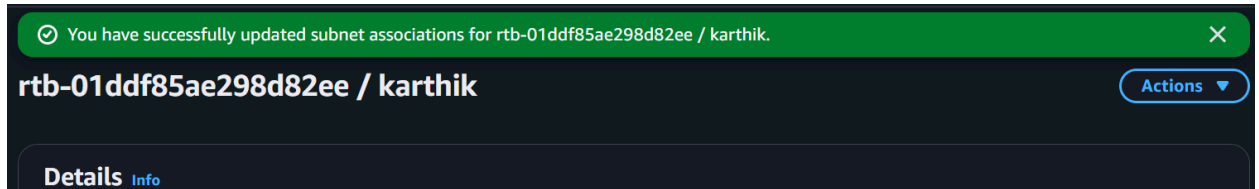
Internet gateways (3) Info Actions Create internet gateway

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID
☐	vinayigw	igw-0836fee5a7c9a8e55	Attached	vpc-00316a204e9280cef vinay
☐	-	igw-0ac23dc0172ab1675	Attached	vpc-076a0f9df0ce8dbe3
☐	karthik	igw-0a9e76539d12757ad	Attached	vpc-0ad023c4fa4cd32c2 karth

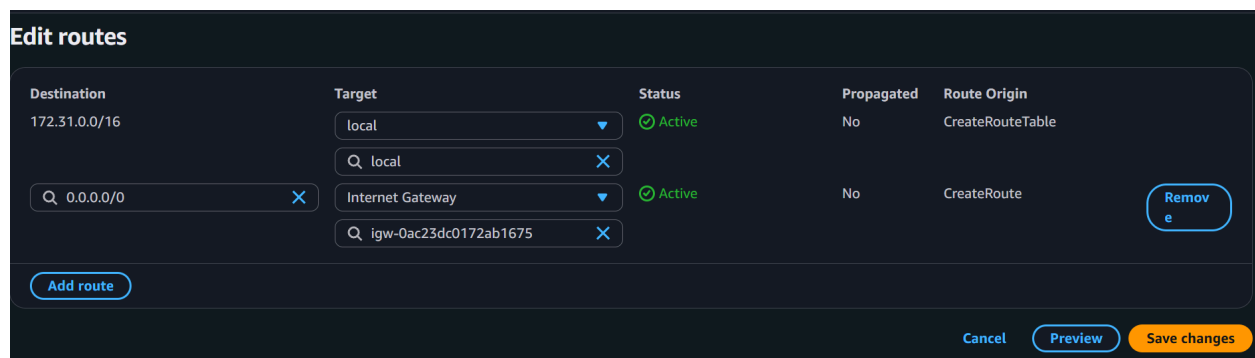
Step 4: Create a Route Table:

- VPC → Route Tables
- Click → Route table
- Name : Public -rt
- Select my-vpc
- Click create



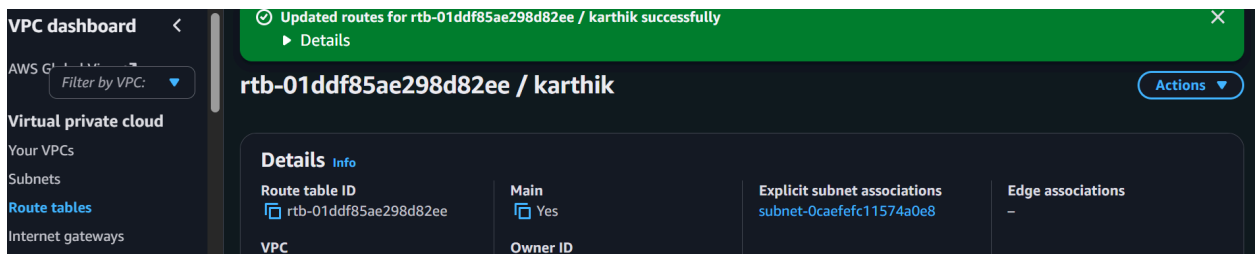
Step 5: Add Internet Route

- Open public-rt.
- Go to Routes tab.
- Click Edit routes.
- Add route:
- Destination: 0.0.0.0/0
- Target: Internet Gateway (my-igw)
- Click Save.



Step 6: Associate Route Table with Subnet:

- In public-rt.
- Go to Subnet Associations.
- Click Edit subnet associations.
- Select public-subnet.
- Click Save



Step 8: Connect to EC2 → After the instance is running:

- Copy the Public IP.
- Open terminal.
- Run it

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17 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-31-198:~$ ping google.com
PING google.com (142.251.167.139) 56(84) bytes of data:
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=1 ttl=105 time=2.04 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=2 ttl=105 time=2.07 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=3 ttl=105 time=2.05 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=4 ttl=105 time=2.04 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=5 ttl=105 time=2.04 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=6 ttl=105 time=2.06 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=7 ttl=105 time=2.04 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=8 ttl=105 time=2.03 ms
64 bytes from ww-in-f139.1e100.net (142.251.167.139): icmp_seq=9 ttl=105 time=2.04 ms
```

